

CHENCHANG WANG

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EDUCATION

University of California, Santa Barbara — B.S. in Computer Science

Expected Mar 2027 | GPA: 3.59

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, Go, JavaScript, TypeScript
- **Backend:** FastAPI, Spring Boot, REST APIs
- **Databases:** PostgreSQL, Redis
- **Systems:** Distributed Systems, Concurrency, Fault Tolerance, System Design
- **Frontend:** React, Next.js, HTML/CSS, Responsive UI
- **AI / Data:** OpenAI API, RAG, PyMuPDF, document chunking
- **Cloud:** Google Cloud (Cloud Run, Compute Engine, VPC, Artifact Registry)
- **Tools:** Git, GitHub, Docker, Linux, JUnit, Playwright

PROJECTS

TaskFlow: Distributed Task Scheduling Platform [\[GitHub\]](#) [\[Live Demo\]](#)

Jul 2026 – Present

Personal Project | Go, PostgreSQL, Redis, React, Docker, Google Cloud

- Built a distributed task scheduler in Go with PostgreSQL-backed state, Redis queues, concurrent workers, and a React dashboard for task and worker monitoring.
- Implemented atomic task claiming, ownership-safe state transitions, bounded worker concurrency, and persistent attempt history to prevent duplicate execution and stale updates.
- Designed retry and failure-recovery workflows with exponential backoff, worker heartbeats, offline detection, timeouts, cancellation, and abandoned-task recovery.
- Deployed the full-stack system on Google Cloud using Cloud Run and a private Compute Engine VM, routing API traffic through VPC networking while keeping backend services off the public internet.

Full-Stack Feature Development — Legacy Code Project

Fall 2025

UCSB Course Project | React, Spring Boot, REST APIs, JUnit, Playwright

- Implemented and maintained full-stack features in an existing production-style codebase by modifying REST APIs and React components.
- Added JUnit and Playwright tests to validate backend behavior and end-to-end user flows.
- Debugged legacy code, investigated API/frontend integration issues, and delivered changes through GitHub pull requests and code reviews.

CoursePilot AI — AI-Powered Course Study Assistant [\[GitHub\]](#)

Summer 2026

Personal Project | Next.js, TypeScript, FastAPI, Python, OpenAI API, PyMuPDF, RAG

- Built a full-stack AI study assistant that converts uploaded course PDFs into study guides, quizzes, reports, and tutoring chat.
- Implemented a RAG pipeline with PDF parsing, document chunking, keyword retrieval, and source-grounded responses.
- Designed quiz scoring, weak-topic analysis, personalized learning recommendations, and AI tutoring workflows based on user performance.
- Implemented multi-turn AI tutoring conversations with configurable response modes, suggested follow-ups, LaTeX rendering, and source-grounded answers.

EXPERIENCE

Intern

Summer 2024

Water Department, Water Authority, China

- Collected and organized building-level geographic data to support flood evacuation planning and emergency response documentation.
- Mapped and numbered residential buildings during fieldwork, improving the accuracy of local disaster-preparedness records.

ADDITIONAL

Languages: Mandarin, English